Tutor: oggi ore 14.30 – 17.30 Coniche

Martedì: simulazione 2ª prova parziale

Mercoledì: Esercizi di ripasso

Venerdì: 2° parziale ore 11.15

$2a_{12}=8$ Caso $a_{12}\neq 0$

$\gamma: \underbrace{x^2+8xy+y^2}-5=0$

$A=\begin{bmatrix}1&4\\4&1\end{bmatrix}$ $\det A=\lambda_1\lambda_2<0 \Rightarrow \gamma$ è iperbole

$A'=\begin{bmatrix}1&4&0\\4&1&0\\0&0&-5\end{bmatrix}$ $\det A'\neq 0 \Rightarrow \gamma$ non degenere

$p(\lambda)=|A-\lambda I|=\begin{vmatrix}(1-\lambda)&4\\4&(1-\lambda)\end{vmatrix}=(1-\lambda)^2-16=\lambda^2-2\lambda-15=0$

$\lambda_1=5 \qquad \lambda_2=-3$

Dobbiamo costruire la matrice $P=[\vec{v}_1,\vec{v}_2]$

t.c. ${}^tP\,A\,P=\begin{bmatrix}5&0\\0&-3\end{bmatrix}$

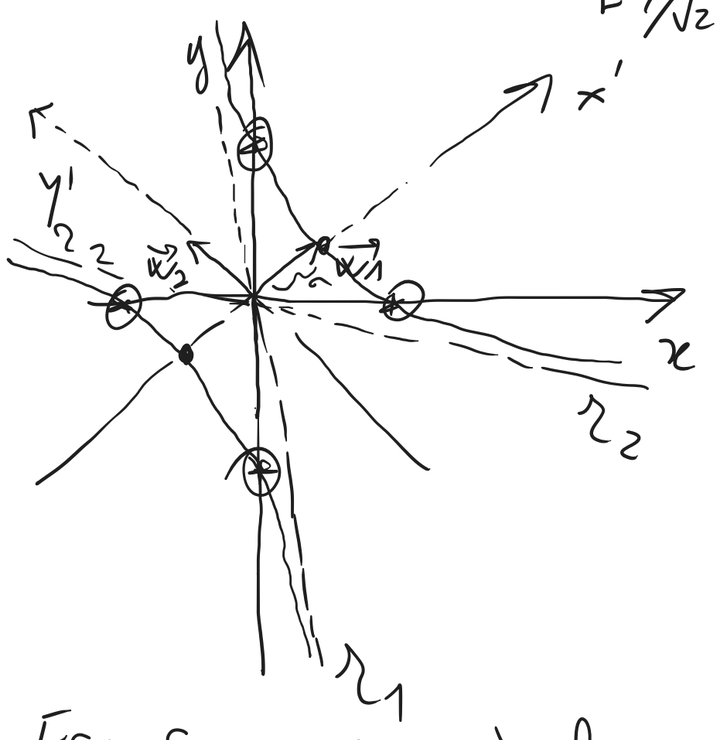
$\vec{v}_1$ sarà il versore di x' ⟶

$\vec{v}_2$ " " " " y'

$\vec{v}_1$ si ricava da $(A-\lambda_1 I)\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$, cioè $\begin{cases}-4x+4y=0\\4x-4y=0\end{cases}$

$A-\lambda_1 I=\begin{bmatrix}-4&4\\4&-4\end{bmatrix}$ $s=1$ $\qquad \vec{v}_1=\text{vers}\begin{pmatrix}1\\1\end{pmatrix}=\begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{bmatrix}$

$\vec{v}_2=\ldots=\begin{bmatrix}-1/\sqrt{2}\\1/\sqrt{2}\end{bmatrix}$ $\qquad P=\begin{bmatrix}1/\sqrt{2}&-1/\sqrt{2}\\1/\sqrt{2}&1/\sqrt{2}\end{bmatrix}$ $\det P=1$ ok



Rispetto a x',y'

$\gamma: x^2+8xy+y^2-5=0$

$\quad \underbrace{\qquad}=\lambda_1 x'^2+\lambda_2 y'^2$

$\gamma: 5x'^2-3y'^2=5$

$\gamma: x'^2-\dfrac{y'^2}{(5/3)}=1$ $\qquad a=1$, $b=\sqrt{\frac{5}{3}}$

$\frac{b}{a}>1$

Asintoti: $r_1: y'=\frac{b}{a}x'$

$r_2: y'=-\frac{b}{a}x'$

Es: Scrivere le eq. degli asintoti r_1 e r_2 rispetto alle coordinate di partenza x,y.

$\begin{bmatrix}x\\y\end{bmatrix}=P\begin{bmatrix}x'\\y'\end{bmatrix} \Rightarrow$

$\begin{bmatrix}x'\\y'\end{bmatrix}={}^tP\begin{bmatrix}x\\y\end{bmatrix}$ cioè $\begin{bmatrix}x'\\y'\end{bmatrix}=\begin{bmatrix}1/\sqrt{2}&1/\sqrt{2}\\-1/\sqrt{2}&1/\sqrt{2}\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}$

$\begin{cases}x'=\frac{1}{\sqrt{2}}(x+y)\\y'=\frac{1}{\sqrt{2}}(-x+y)\end{cases}$

$r_1: y'=\frac{b}{a}x'$

$r_2: y'=-\frac{b}{a}x'$

$r_1: \frac{1}{\sqrt{2}}(-x+y)=\frac{b}{a}\frac{1}{\sqrt{2}}(x+y)$

r_2 simile